LINUX ESSENTIALS
CHAPTER 2
GETTING HELP.

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Objective

- In this lab, you will perform the following tasks:
  - Use several help systems to get help for commands.
  - Learn how to locate commands.
- In this chapter you will explore the how to get help. This will be a very useful thing to know how to do when you find yourself stuck or when you can't remember how a command works.
- In addition to Internet searches, the Linux Operating System provides a variety of techniques to learn more about a given command or feature. Knowing these different techniques will allow you to more easily and quickly find the answer you need.

GETTING HELP

- In this task, you will explore the how to get help. This will be a very useful thing to know how to do when you find yourself stuck or when you can't remember how a command works.

- In addition to Internet searches, the Linux Operating System provides a variety of techniques to learn more about a given command or feature. Knowing these different techniques will allow you to more easily and quickly find the answer you need.

Getting Help

- To learn more about commands, access the manual page for the command with the man command.
- You can move through manual page using characters in this table.
- Note:
  - Search in manual page not case sensitive.

```
sysadmin@localhost:~$ date
Tue Jan 19 17:27:20 UTC 2016
sysadmin@localhost:~$
```

```
sysadmin@localhost:~$ man date
```

Key	Purpose
H or **h**	Display the help
Q or **q**	Quit the help or manual page
Spacebar or **f** or **PageDown**	Move a screen forward
b or **PageUp**	Move a screen backward
Enter or **down arrow**	Move down one line
Up arrow	Move up one line
/ followed by text to search	Start searching forward
? *followed by text to search*	Start searching backward
n	Move to next text that matches search
N	Move to previous matching text

Getting Help

- The passwd command used to change password for current user.
- In some cases you may not remember the exact name of the command. In these cases you can use the -k option to the man command and provide a keyword argument.
- The -k option to the man command will often produce a huge amount of output.
- We can use apropos instead of man –k.

```
sysadmin@localhost:~$ man -k password
chage (1)            - change user password expiry information
chgpasswd (8)        - update group passwords in batch mode
chpasswd (8)         - update passwords in batch mode
cpgr (8)             - copy with locking the given file to the pa
cppw (8)             - copy with locking the given file to the pa
expiry (1)           - check and enforce password expiration poli
login.defs (5)       - shadow password suite configuration
pam_pwhistory (8)    - PAM module to remember last passwords
pam_unix (8)         - Module for traditional password authentica
passwd (1)           - change user password
passwd (1ssl)        - compute password hashes
passwd (5)           - the password file
pwck (8)             - verify integrity of password files
pwconv (8)           - convert to and from shadow passwords and g
shadow (5)           - shadowed password file
shadowconfig (8)     - toggle shadow passwords on and off
unix_chkpwd (8)      - Helper binary that verifies the password o
unix_update (8)      - Helper binary that updates the password of
vipw (8)             - edit the password, group, shadow-password
sysadmin@localhost:~$
```

Getting Help

- There are often multiple man pages with the same name. For example, the previous command showed three pages for passwd.
- there are different man pages for the same "name" is confusing for many beginning Linux users.
- A command such as man passwd, the first section is searched and, if a match is found, the man page is displayed.
- The man -f passwd command that you previously executed shows that there is a section 1 man page for passwd: passwd (1).

```
sysadmin@localhost:~$ man -f passwd
passwd (5)           - the password file
passwd (1)           - change user password
passwd (1ssl)        - compute password hashes
sysadmin@localhost:~$
```

The different man pages are distinguished by "sections". By default there are nine default sections of man pages:

1. Executable programs or shell commands
2. System calls (functions provided by the kernel)
3. Library calls (functions within program libraries)
4. Special files (usually found in `/dev`)
5. File formats and conventions, e.g. `/etc/passwd`
6. Games
7. Miscellaneous (including macro packages and conventions), e.g. `man(7)`, `groff(7)`
8. System administration commands (usually only for root)
9. Kernel routines [Non standard]

Getting Help

- Instead of man –f passwd we can use whatis passwd.

- To display a man page for a different section,
  - provide the section number as the first argument to the man command.

```
man 5 passwd
```

```
PASSWD(5)                File Formats and Conversions

NAME
       passwd - the password file

DESCRIPTION
       /etc/passwd contains one line for each user account, with
       delimited by colons (":"). These fields are:

       o   login name

       o   optional encrypted password

       o   numerical user ID

       o   numerical group ID

       o   user name or comment field

       o   user home directory

       o   optional user command interpreter

 Manual page passwd(5) line 1 (press h for help or q to quit)
```

Getting Help

- Almost all system features (commands, system files, etc.) have man pages.
- Some of these features also have a more advanced feature called info pages.
- Many beginning Linux users find info pages to be easier to read. They are often written more like "lessons" while man pages are written purely as documentation.

```
info date
```

```
File: coreutils.info,  Node: date invocation,  Next: arch invocation,  Up:
em context

21.1 `date': Print or set system date and time
=============================================

Synopses:

     date [OPTION]... [+FORMAT]
     date [-u|--utc|--universal] [ MMDDhhmm[[CC]YY][.ss] ]

   Invoking `date' with no FORMAT argument is equivalent to invoking it
with a default format that depends on the `LC_TIME' locale category.
In the default C locale, this format is `'+%a %b %e %H:%M:%S %Z %Y'',
so the output looks like `Thu Mar  3 13:47:51 PST 2005'.

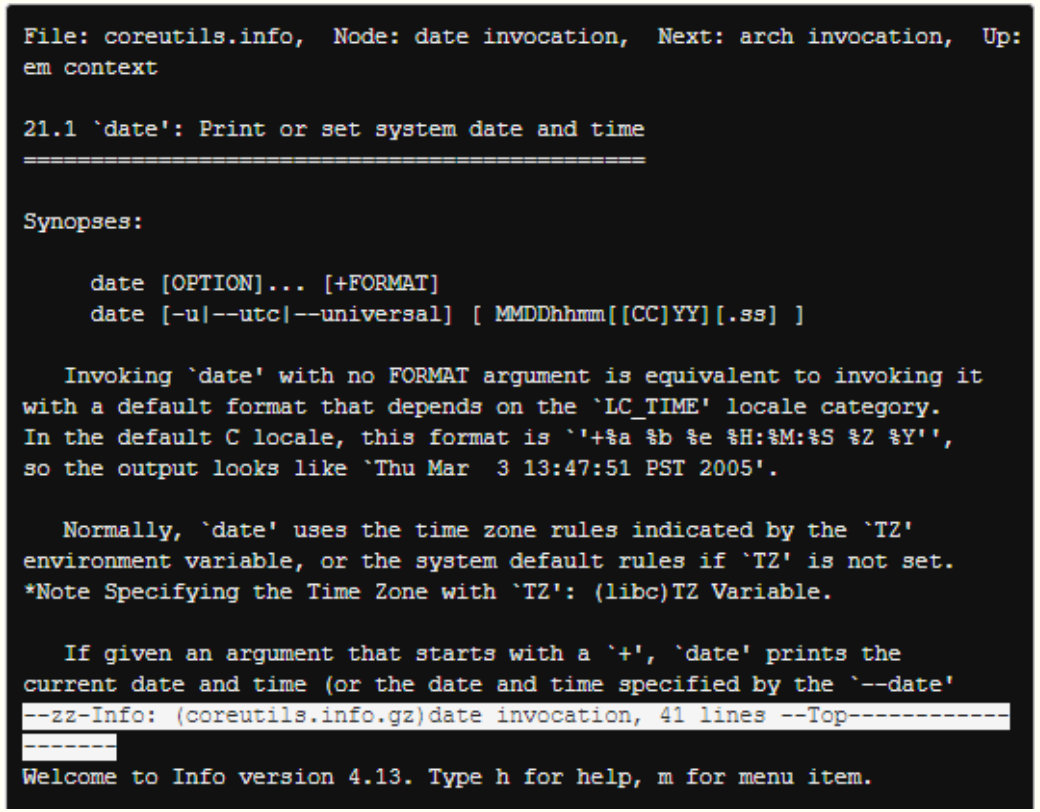
   Normally, `date' uses the time zone rules indicated by the `TZ'
environment variable, or the system default rules if `TZ' is not set.
*Note Specifying the Time Zone with `TZ': (libc)TZ Variable.

   If given an argument that starts with a `+', `date' prints the
current date and time (or the date and time specified by the `--date'
--zz-Info: (coreutils.info.gz)date invocation, 41 lines --Top-----------
-------
Welcome to Info version 4.13. Type h for help, m for menu item.
```

Getting Help (info)

- Almost all system features (commands, system files, etc.) have man pages.
- Some of these features also have a more advanced feature called info pages.
- Many beginning Linux users find info pages to be easier to read. They are often written more like "lessons" while man pages are written purely as documentation.
- Type the letter **h** to see a list of movement commands.
- After reading the movement commands, type the letter **l** (lowercase L) to return to viewing the document.

```
info date
```



Getting Help (--help)

- Another way of getting help is by using the --help option to a command.
- Most commands allow you to pass an argument of --help to view basic command usage:
- Some system features also have more detailed help documents located in the /usr/share/doc directory structure.
- Note that in almost all cases,
  - the man pages and info pages will provide you with the information that you need.
  - However, if you need more in-depth information (something that system administrators sometimes need), then you may find this information in the files located in the /usr/share/doc directory.

```
date --help
```

```
sysadmin@localhost:~$ date --help
Usage: date [OPTION]... [+FORMAT]
  or:  date [-u|--utc|--universal] [MMDDhhmm[[CC]YY][.ss]]
Display the current time in the given FORMAT, or set the system date.

  -d, --date=STRING         display time described by STRING, not `now'
  -f, --file=DATEFILE       like --date once for each line of DATEFILE
  -r, --reference=FILE      display the last modification time of FILE
  -R, --rfc-2822            output date and time in RFC 2822 format.
                            Example: Mon, 07 Aug 2006 12:34:56 -0600
      --rfc-3339=TIMESPEC   output date and time in RFC 3339 format.
                            TIMESPEC=`date', `seconds', or `ns' for
                            date and time to the indicated precision.
                            Date and time components are separated by
                            a single space: 2006-08-07 12:34:56-06:00
  -s, --set=STRING          set time described by STRING
  -u, --utc, --universal    print or set Coordinated Universal Time
      --help     display this help and exit
      --version  output version information and exit
```

```
ls /usr/share/doc
```

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```

```
ls /usr/share/doc
```

FINDING FILES

- In this task, we will explore how to search for a file on the system. This is useful to know in situations when you can't find a file on the system, either one that you created or one that was created by someone else.

Finding Files

- An easy way to search for a file is to use the locate command.
- Note that
  - the output from the previous example includes files that have crontab as part of their name.
- To find files that are just named crontab use locate –b "\crontlab"

```
sysadmin@localhost:~$ locate crontab
/etc/crontab
/usr/bin/crontab
/usr/share/doc/cron/examples/crontab2english.pl
/usr/share/man/man1/crontab.1.gz
/usr/share/man/man5/crontab.5.gz
sysadmin@localhost:~$
```

```
sysadmin@localhost:~$ locate -b "\crontab"
/etc/crontab
/usr/bin/crontab
sysadmin@localhost:~$
```

Finding Files

- You may just want to find where a command (or its man pages) is located. This can be accomplished with the whereis command .

```
sysadmin@localhost:~$ whereis passwd
passwd: /usr/bin/passwd /etc/passwd /usr/share/man/man1/passwd.1.gz /usr/sh
sysadmin@localhost:~$
```

- Note that:
  - The whereis command only searches for commands and man pages, not just any file.
- Recall from earlier that there is more than one passwd man page on the system. This is why you see multiple file names and man pages (the files that end in .gz are man pages) when you execute the previous command.


Thank
you